Subject Erosion as Social Reproduction: Education, Depressive Fixation, and the Conditions for Subject Protection

Child Paper 5 — English Full Text v1.0 (Complete Academic Version)

Author: Yuki Tanaka (田中佑樹) **Affiliation:** Independent Researcher, Seto Inland Sea Region, Japan **Email:** garamist@dream.jp **ORCID:** https://orcid.org/0009-0002-6183-7886 **Correspondence:** garamist@dream.jp

Keywords: social reproduction, designed simplification of evaluative space, second pathway, Region B, depressive promotion, subject protection, four principles, multiplicative structure

---

Abstract

This paper describes subject-injury not as a malfunction internal to individuals but as a dynamic reproduced by institutional design. Building on the minimal causal chain formalized in Child Paper 2, it introduces a *second pathway* into subject-injury — the designed simplification of evaluative space — which operates independently of rising conformity pressure and can deplete D (difference-retention) through the reduction of the evaluative axis to a single dimension alone.

The paper develops this claim across three domains. In education, single-axis evaluation systems are shown to structurally mass-produce Region B: maintaining cognitive autonomy T_cog at a high level while eroding evaluative autonomy T_eval and value autonomy T_val, such that high achievers may be positioned closest to Region B. In the clinical domain, the paper proposes a structural connection hypothesis: T_val fixation — in which value frameworks cease to update while the subjective sense of autonomy is preserved — can form a mutually amplifying loop with depressive promotion, mediated by institutional design. In the institutional domain, subject protection is formalized not as an ethical add-on but as a dynamic necessary condition for sustaining a recomputable society, whose minimum requirements are expressed as four rights (right to delay, right to dissent, right to withdraw, right to non-optimization) and translated into four principles of institutional design.

The four principles form a multiplicative structure: the absence of any one renders the others functionally inoperative. The paper concludes that subject-injury persists not by breaking down but by appearing to function well, and that subject protection is therefore the institutional condition for detecting — and interrupting — the cessation that wears the face of success.

A note on the theoretical frameworks

OMD (Organizational Metastability Dynamics), RD (Recomputation Dynamics), and RLAF (Reinforcement Learning from AI Feedback) are theoretical frameworks developed within the present research program and are introduced here as analytical constructs rather than references to previously published models under those names.

---

1. Introduction

1.1 The Problem

Why can something that looks like success be subject-injury?

In the context of education, a student who earns better grades, completes assignments, and achieves entrance targets is regarded as having "succeeded." In the clinical context, a patient in a depressive state is described in terms of symptoms: declining motivation and low self-evaluation. In the institutional context, a lack of subjectivity is readily treated as a problem of individual capacity or character.

But these observations may simply be different cross-sections of the same dynamic, viewed in isolation from one another.

The preceding paper (Child Paper 2) showed that subject-injury can proceed while surface functioning is preserved. In Region B — apparently normal, internally halted — the state in which evaluative autonomy T_eval has become paralyzed and value autonomy T_val has become fixed, cognitive autonomy T_cog remains, so output continues. Grades are maintained, responses are stable, and the problem is not visible on the surface. But internal updating has already ceased.

This makes it possible to pose the same question across three domains — education, clinical care, and institutional design:

> Is the process by which internal updating ceases while apparent normality is maintained not an incidental internal event within individuals, but something that is structurally produced by institutional design?

The layer that existing research misses

Educational research asks how to support learner agency and promote autonomy. Clinical research asks about the description of depressive symptoms and the effects of intervention. Institutional theory asks about rights design and policy implementation. Each of these has accumulated important findings — but they are structurally disconnected from each other.

As a result, a layer is being missed. The same dynamic — the depletion of D (difference-retention) and the cessation of recomputation capacity M — is being repeatedly reproduced across the distinct domains of education, depressive promotion, and institutional design. Making this visible requires a unified descriptive framework that shares the same dynamic core.

The position of this paper

This paper formalizes this dynamic as follows:

> Subject-injury is not a malfunction internal to individuals but a dynamic reproduced by institutional design.

When institutions are designed in a specific way — when evaluative axes are reduced to a single dimension, when suspension is excluded as inefficient, when difference is processed as noise — the dynamics of subject-injury are structurally activated. Well-intentioned institutions can mass-produce Region B.

What this paper inherits from Child Paper 2

This paper takes the dynamics formalized in Child Paper 2 as its given premises. The definitions of Region B, the minimal causal chain, and D/M/T_eval/T_val are not revisited here.

The single theoretical point that this paper adds to Child Paper 2's dynamics is the second pathway: the designed simplification of evaluative space. In this pathway, even when C is low, D can be depleted. From this starting point, the three distinctive developments of this paper follow: a description of the mechanism by which educational institutions mass-produce Region B; the connection hypothesis linking T_val fixation to depressive promotion; and the formalization of the four principles of subject protection.

Organization of this paper

Section 2 extends Child Paper 2's dynamics into a theory of institutional reproduction. Section 3 describes the mechanism by which educational institutions mass-produce Region B and formalizes the conditions for intervention. Section 4 cross-references the connection hypothesis linking T_val fixation to depressive promotion against existing research. Section 5 formalizes the institutional conditions for subject protection as a multiplicative structure. Section 6 discusses the significance and limits of the integration. Section 7 presents the conclusion.

1.2 Position Relative to Existing Research

The achievements and limits of educational research

Research on learner agency, autonomy support, and self-regulated learning has expanded rapidly since 2024, accelerated by the spread of generative AI. Questions about whether AI supports or undermines learner subjectivity, the conditions for self-directed learning, and the redefinition of learner agency have all produced important practical findings.

These bodies of research, however, operate within a support-oriented framework. Subject-injury that presents under the appearance of success cannot be diagnosed from a support-oriented perspective. The question of whether institutions are reproducing subject-injury is not one that can be posed from within that framework.

The achievements and limits of clinical and psychological research

Research on self-determination theory and learned helplessness has illuminated the relationship between autonomy and intrinsic motivation, and the process by which uncontrollable situations undermine agency. These bodies of research, however, operate within a framework that describes depression primarily as an individual psychological state. The kind of dynamic description found in T_val fixation — the cessation of recomputation — lies outside that framework.

The achievements and limits of institutional design and rights theory

Rights theory provides an important normative foundation. But it does not describe why those rights are necessary in terms of dynamics. As long as the basis remains normative rather than dynamic, both falsification and design improvement remain difficult.

The novelty of this paper and its points of connection

The structure in which the depletion of D and the cessation of M are repeatedly produced across the three domains of education, clinical care, and institutional design has not yet been described as a single dynamic.

This paper attempts to fill this gap, building on the dynamics formalized in Child Paper 2. Specifically, it adds three elements. First, the second pathway of subject-injury: the designed simplification of evaluative space, independent of rising conformity pressure C. Second, a structural account of the connection hypothesis linking T_val fixation to depressive promotion. Third, the translation of the dynamic conditions — the institutionalization of D, H, and recomputability — into the four principles of subject protection.

A note on vocabulary

The vocabulary used in this paper — Region B, subject-injury, D, T_eval paralysis, and so on — should not be used as authoritative labels for making unilateral determinations about others from the outside. The primary use of this vocabulary is the self-audit of institutional, environmental, and evaluative design: analytical concepts for designers to examine whether their own designs are reinforcing the reproduction cycle of subject-injury.

Furthermore, this paper does not claim D (difference-retention) as an unconditional good. It does not deny that there are contexts in which homogenization and standardization function effectively. D is positioned as a necessary condition only in contexts where the functional requirement at issue is whether subjects remain capable of recomputation in response to environmental change.

---

2. Subject Erosion as Social Reproduction

2.1 The Designed Simplification of Evaluative Space

The pathway in Child Paper 2 and its limits

Child Paper 2 described a pathway in which the rise of C drives the decline of D. This account, however, presupposes that the rise of C — a pressure from outside — comes first.

There are, however, situations in which D is depleted even when C is low. In such situations, subject-injury can proceed quietly without any external coercion, without any force that could be called conformity pressure. This paper calls that pathway the designed simplification of evaluative space.

What evaluative space is

Evaluative space refers to the set of criteria that a system consults when judging "what constitutes a good state" or "what constitutes success." The breadth or narrowness of evaluative space can change independently of the level of C. It happens quietly, through institutional design.

The definitional difference between C and the designed simplification

C is defined as the dynamic pressure applied to a subject through interaction in the relational field. The designed simplification of evaluative space, by contrast, refers to a condition in which the space of evaluative criteria is statically designed to be narrow before interaction even begins. The former is a problem of relational dynamics; the latter is a structural problem of institutional design. The two are independent pathways — the designed simplification can advance even in a system where C is low.

Put differently: C is a force that "someone pushes," while the designed simplification is a condition in which "the space for movement is narrow from the start."

The four pathways of designed simplification

First, concentration on measurable indicators. Criteria that are difficult to measure — the retention of a sense of incongruity, the reframing of a question, the pursuit of alternative approaches — are institutionally invalidated.

Second, the reduction of success conditions to a single line. The cost of maintaining alternative pathways rises, and the space of alternatives e contracts in practice.

Third, the prioritization of speed. Suspension, deliberation, and reconsideration are treated as inefficient, and the room for H (suspension capacity) to operate is by design reduced.

Fourth, the institutional rise in the cost of deviation. The cost of maintaining alternative interpretations, minority views, or reconsideration manifests as invalidation — "not evaluated," "not recorded."

When these four pathways operate simultaneously, $\mathrm{Var}(\varphi)$ tends to contract substantially under institutional design, e becomes impoverished, and the operating preconditions of T_{eval} are readily lost.

The core of the second pathway

> The designed simplification of evaluative space is a pathway into subject-injury independent of rising conformity pressure C. In this pathway, even when C is low, the convergence of evaluative criteria depletes D (difference-retention), impoverishes the exploratory dimension e, and causes the operating preconditions of T_{eval} to be lost.

2.2 Subject-Injury as Social Reproduction

What social reproduction means here

"Social reproduction" in this paper refers to the process by which the same evaluative structure is repeated across generations, groups, and cycles of institutional operation, and is self-reinforced by the institution's own success indicators. No one designs an institution with the purpose of producing subject-injury. Yet as a consequence of design, evaluative space that promotes subject-injury is repeatedly supplied. This "reproduction without intent" is the object of description in this paper.

Why simplified evaluative space is maintained

A single evaluative axis is easy to measure, easy to compare, and easy to distribute responsibility across. These are genuine institutional rationalities. But that optimization does not include the maintenance of D and the continuation of M among its evaluative objects.

Why it appears as success

Subjects raised within a simplified evaluative space maintain high T_cog output, sustain σ, and continue to respond to the required indicators. The institution misreads this state as "success." But as Child Paper 2 showed, this state is the prototype of Region B. Because the problem is not visible, the institution is not corrected. Because it is not corrected, the same evaluative space is supplied to the next cohort.

Why it is therefore reproduced

High σ and T_cog output are recorded as "excellent results." The institution invests further resources in the same design, refines the evaluative axis further, and optimizes speed further. This produces a deeper Region B.

> The institution misreads Region B as achievement, reinforces its design on the basis of that misreading, and reproduces a deeper Region B. This is the mechanism of the social reproduction of subject-injury.

---

3. Education as a Region B Production System

3.1 Single-Axis Evaluation Systems

What does a single-axis evaluation system optimize?

The term "single-axis evaluation system" is used here not as a value judgment but as a structural description. A single-axis evaluation system refers to the institutional design of education that records and compares relative academic ranking in a measurable form and operates that ranking as the primary selection criterion.

The analysis in this paper applies not only to any particular national system, but to any educational institution that operates a measurable single evaluative axis as its primary selection criterion in general — including selection systems centered on standardized tests and progression decisions based on GPA as a single indicator.

The problem is not the evaluative axis itself, but what happens when it becomes the sole or overwhelmingly dominant axis of the evaluative space.

When a single evaluative axis operates

When a single evaluative axis becomes dominant, the four pathways described in Section 2.1 operate simultaneously.

Concentration on measurable indicators: the knowledge, skills, and processing speed measurable by academic tests and standardized assessments become the objects of evaluation. Criteria that are difficult to measure — the retention of a sense of incongruity, the reframing of a question, the pursuit of alternative approaches — are institutionally invalidated.

Reduction of success conditions to a single line: speed and accuracy in arriving at correct answers become the success conditions for selection. Exploring alternative pathways is treated as a waste of time. The space of alternatives e contracts.

Prioritization of speed: answering many questions correctly within a time limit is valued. Suspension, deliberation, and reconsideration become grounds for deduction. The room for H (suspension capacity) to operate is by design reduced.

Institutional rise in the cost of deviation: responses that do not conform to scoring criteria are processed as “incorrect answers.” Alternative framings, questions about premises, and minority interpretations appear as invalidation — “not evaluated,” “not recorded,” “without effect on academic progression.”

When these four pathways operate simultaneously, Var(φ) tends to contract substantially under institutional design, e becomes impoverished, and the operating preconditions of T_eval are readily lost.

Why high achievers may be positioned closest to Region B

Learners who maintain high achievement under a single evaluative axis have most successfully optimized for that axis. At the same time, however, this optimization most efficiently advances the depletion of D. T_eval is optimized in the direction of “eliminating difference as noise.”

When opportunities to convert difference inputs — inputs capable of prompting re-evaluation of the existing frame — into recomputation resources diminish, T_eval becomes unable to supply new evaluative input to T_val. As a result, T_val tends to fix the existing success criteria as values in themselves.

From the outside, the subject appears to be succeeding. But internally, the transition toward Region B may be proceeding most rapidly.

This is a structural tendency claim about institutional design. Subjects positioned at the point where the optimization pressure toward a single evaluative axis is strongest are most readily deprived of D through designed simplification.

The difficulty of diagnosis

The residual of T_cog renders Region B invisible. Since measurable output is maintained, diagnostic indicators show no abnormality. The problem is proceeding in a layer that the indicators are not measuring — the paralysis of T_eval and the fixation of T_val.

3.2 Formalizing the Conditions for Intervention

Why individual effort is insufficient for intervention

The design of a single evaluative axis simultaneously depletes D and H through four pathways. To demand of individuals within this structure that they "retain difference," "stop and pause," and "reconsider" is isomorphic to demanding that they stop in a space designed for movement. The demand is correct, but the design does not permit it.

Intervention must therefore be directed at the design of the evaluative space itself.

Formalizing the three conditions

Condition 1: Two-tier evaluation design

Achievement evaluation is maintained as it is. In addition, a layer of recomputation evaluation is institutionally established. What recomputation evaluation measures is whether the premises of the question can be rephrased, whether the limits of a given solution method can be described, and whether an alternative evaluative axis can be independently proposed. What is important in the two-tier design is that the second tier functions as an independent evaluative layer.

Condition 2: Institutional protection of H (suspension capacity)

Not giving an immediate response is not penalized. Holding an answer in suspension is treated as a valid response. "Not yet decided" is recorded as a legitimate intermediate state. "Institutionalized time for reconsideration" is positioned as a concrete implementation of this condition.

Condition 3: Including difference-retention in the success indicators

Whether minority views remain, whether alternative solutions are on record, whether objection is possible — these are made indicators for evaluating $\mathrm{Var}(\varphi)$ in the relational field of the classroom, school, and institution. When institutions include the maintenance of difference in their self-evaluation, the basis for interrupting the self-reinforcing cycle of designed simplification is established.

The multiplicative structure of the three conditions

These three conditions form a multiplicative structure. Without two-tier evaluation, the operation of T_{eval} is not rewarded. Without the protection of H, reconsideration is not executable. Without difference as a success indicator, the institution returns to a single evaluative axis. This is structurally analogous to the $A(e,u) = f(e)\times g(u)$ structure demonstrated in Child Paper 2.

Partial implementation can yield local improvements, but the simultaneous satisfaction of all three is a necessary condition for structural stability in subject protection.

---

4. Subject Erosion and Depressive Fixation

4.1 The Connection Hypothesis: T_val Fixation and Depressive Fixation

The position of this section

This section does not perform medical diagnosis. The fixation of T_val is not a sufficient condition for depressive promotion, but it may function as a structural condition promoting depressive promotion in environments where the cessation of recomputation persists.

What T_val fixation is

The fixation of T_val begins with the paralysis of T_eval. When T_eval becomes unable to convert difference inputs — inputs capable of prompting re-evaluation of the existing frame — into recomputation resources, the supply of updating input to T_val is cut off. The longer this maintenance continues, the more the divergence from the external environment accumulates; but the circuit that would sense this divergence (T_eval) is already not functioning.

Points of connection and difference from existing research

Self-determination theory (Deci & Ryan) showed that the absence of autonomy is associated with the decline of intrinsic motivation and the promotion of depression. T_val fixation has a point of connection with this account.

The difference, however, must be made explicit. What self-determination theory is concerned with is the decline in the subjective sense of autonomy — the subject feels “I am not deciding for myself.” T_val fixation is different. A subject whose T_val has become fixed continues to feel “I am acting in accordance with my own values,” while the state is one in which those values are no longer being updated. The fact that the subjective sense of autonomy is preserved while internal updating has ceased is the distinctive feature of T_val fixation, and its consistency with Region B.

In terms of the analogy to learned helplessness (Seligman): when a state continues in which updating is attempted but the environment does not respond, updating itself is abandoned. However, the difference that helplessness is an outcome while T_val fixation is its structural condition is maintained.

Why T_val fixation can become a structural condition for depressive promotion

The subject repeatedly experiences their own actions as having no effect on the environment. But the fact that the cause of this “having no effect” lies in the divergence at the level of the frame cannot be sensed.

Because the subject cannot identify the cause as lying in the divergence on the frame side, the response-mismatch that results tends to be received as the subject’s own ineffectiveness or the meaninglessness of their actions. The repeated experience of “nothing changes no matter what I do” is therefore structurally produced in a system where the frame is not updated. This can function as a structural condition promoting depressive promotion.

The scope and limits of the hypothesis

First, causal relations are not demonstrated in this paper. Second, depression is multifactorial. Third, individual differences are abstracted away.

Summary

The fixation of T_val — the state in which the value frame ceases to update while the subjective sense of autonomy is preserved — structurally produces a sense of the ineffectiveness of action through the inability to sense divergence, and can function as a structural condition promoting depressive promotion. The following section describes how single-axis evaluation, speed prioritization, and the inability to suspend collectively sustain T_val fixation and constitute the institutional mediation of depressive promotion.

4.2 Depressive Promotion as Institutional Mediation

Why do institutions sustain T_val fixation?

Even when T_val fixation has occurred once within an individual, there is in principle the possibility of returning to recomputation through the experience of incongruity with the external environment. The problem arises when the pathway back to recomputation is closed by institutional design.

In a single-axis institution, incongruity with the environment tends to be interpreted as "a problem of my own insufficient adaptation." Fixed T_val is externally reinforced by the institution. In a speed-prioritizing institution, there is no time to slowly review divergence. The inability to suspend locks both of these in place. When H is lost, frame-level updating equivalent to L3 does not activate.

The dynamics of institutional mediation

As shown in Section 4.1, this mediation can obtain even without the subject feeling suppressed by the institution. The more rational, measurable, and fair the institution appears, the stronger this mediation becomes.

It is sufficient for the institution to unify its success indicators, reward subjects who respond quickly to those indicators, and not evaluate suspension. In this state, the subject feels they are voluntarily maintaining their own T_val, while in reality they are circulating within a fixed frame reinforced by the institution.

The cause of the mismatch tends to be attributed not to the institution but to the self. In a system where T_eval is paralyzed, the question tends to be absorbed into self-attributed lack of capacity or lack of effort. This is where the institutional mediation of depressive promotion can become established.

Depressive promotion as a structural condition

Depressive promotion can be read not merely as individual pathology but as a consequence of the evaluative space supplied by institutional design. It is possible to describe the dynamic in which institutions mediate the cessation of recomputation, and that cessation sustains the promoting conditions for depressive promotion.

Summary

This paper does not claim a direction of causation for depressive promotion. What it claims is the structural hypothesis that T_val fixation and depressive promotion can form a mutually amplifying loop. When T_val becomes fixed, the divergence from the environment becomes imperceptible, and the sense of the ineffectiveness of action accumulates. The accumulated sense of ineffectiveness further reduces the impetus for recomputation, further reinforcing T_val fixation. In institutional environments where this cycle obtains, depressive promotion can arise from within the interaction between the institutional environment and the fixed evaluative frame. Whether the direction of causation runs from T_val fixation to depressive promotion, or from depressive promotion to T_val fixation, remains a question requiring longitudinal empirical study.

---

5. Conditions for Subject Protection

5.1 Why Subject Protection Is a Necessary Condition

What the preceding chapters have established

The designed simplification of evaluative space depletes D through a pathway independent of rising conformity pressure C. That depletion is maintained and reinforced by institutional rationality, repeatedly producing Region B. Educational institutions are the prototypical implementation of this reproduction cycle, most readily positioning high achievers closest to Region B. The fixation of T_val, reinforced by institutional design, accumulates divergence from the environment, structurally repeats the sense of the ineffectiveness of action, and can establish a mediating pathway for depressive promotion.

Faced with this chain, the question becomes: where, by whom, and what is to be protected?

Why subject protection is not placed as an ethical slogan

As long as subject protection is positioned as “ethically desirable,” it will continue to be placed at the bottom of the institutional design priority list. The position of this paper is different. Subject protection is not something done when resources permit — it is the minimum condition for sustaining a recomputable society.

Why it is a “necessary condition”

Region B forms a self-sustaining cycle. The cessation of M causes a further decline of D, which depletes E_r_eff, making the reactivation of M still more difficult. Unless conditions are established to interrupt this cycle from outside, institutions will continue to mass-produce Region B while reinforcing themselves.

Subject protection is the institutional arrangement for interrupting this cycle from outside. It is not an ethical add-on. It is the minimum requirement for institutions to maintain their capacity for self-renewal.

The dynamic basis

The three dynamic conditions — replenishment of D, recovery of the conditions for recomputation, and securing H — form a multiplicative structure. This is not to deny incremental improvement as such, but partial implementation of only one of the three conditions will, given the nature of the multiplicative structure, not function structurally as subject protection.

Recovery of the conditions for recomputation here refers to the institution providing subjects with the margin and resource conditions sufficient to activate recomputation; at the institutional design level of this paper, it is implemented primarily through the securing of H and the two-tier design of evaluative space. This establishes the bridge between the three conditions at the dynamic level and the four principles at the institutional level.

Organization of this chapter

Section 5.2 translates the three conditions — the maintenance of D, the securing of H, and the institutionalization of recomputability — into the institutional language of the four rights: the right to delay, the right to dissent, the right to withdraw, and the right to non-optimization. The right to delay corresponds to the institutional recognition of H; the right to dissent corresponds to the institutional maintenance of D; the right to withdraw and the right to non-optimization correspond to the conditions for suppressing over-optimization toward a single evaluative axis. Section 5.3 presents the minimum principles for embedding these in institutional design.

5.2 The Dynamic Formalization of the Four Rights

The logic of translation

The four rights are a translation of the three conditions into institutional language. Rights language expresses the requirements of institutional design in a form that subjects can demand from their own side.

Among the three conditions, “the institutionalization of recomputability” does not correspond to any single one of the four rights; it is a condition that can only be established when all four rights function simultaneously as a multiplicative structure.

The right to delay — the institutional recognition of H (suspension capacity)

The right to delay is the right not to be compelled to respond immediately. Without H, even when difference is supplied, recomputation will not activate. By institutionally recognizing “not yet decided,” “reframing the question,” and “stopping once” as legitimate responses, the room for H to operate is embedded in the institution.

The right to withdraw — the right to temporarily distance oneself from the dominant evaluative axis

Whereas the right to delay is the right to stop within the evaluative process, the right to withdraw is the right to temporarily distance oneself from the dominant evaluative axis itself. The former is an intervention on the temporal axis of evaluation; the latter is an intervention on the spatial axis, and the two are distinguished accordingly.

A single evaluative axis institutionally raises the cost of deviation and depletes D. The right to withdraw is the securing of the space to reconsider one's relationship to the evaluative axis.

The right to dissent — the institutional maintenance of D (difference-retention)

The right to dissent is the right not to have difference processed as noise. When difference is "not evaluated" or "not recorded," D becomes too costly to maintain and is depleted. The right to dissent is the institutional protection that enables difference to continue to exist in the evaluative space.

The right to non-optimization — the right to refuse to surrender the evaluative core

The right to non-optimization is the right not to be required to surrender one's own evaluative frame to the institution on grounds of efficiency or consistency. It is positioned as the requirement for securing a certain distance from institutionally enforced optimization, in order to keep T_val in a state capable of updating. The right to non-optimization is therefore understood not as a refusal of achievement or responsibility, but as an institutional condition for not fully identifying the value frame of the subject with the institution's success criteria.

The mutual dependence and multiplicative structure of the four rights

Without the right to delay, dissent cannot be expressed. Without the right to dissent, withdrawal becomes meaningless. Without the right to withdraw, non-optimization cannot be sustained. Without the right to non-optimization, if the institution converges T_val toward a single success criterion, delay, dissent, and withdrawal are all absorbed as preparation for entry into the institution's evaluative axis.

The four rights stand in a multiplicative structural relationship. The absence of any one of them can render the other three functionally inoperative in practice. Partial implementation can yield local improvements, but the simultaneous functioning of all four rights is a necessary condition for structural stability as subject protection.

5.3 The Minimum Principles of Institutional Design

The position of this section

Section 5.2 formalized the four rights. Section 5.3 presents the minimum principles for embedding those four rights in actual institutional design.

Principle 1: Two-tier evaluation design

This principle is the design condition corresponding to both the right to non-optimization and the right to dissent as formalized in Section 5.2. If the right to non-optimization protects the subject's evaluative core, the two-tier design is the form in which the institution structurally guarantees that protection.

The first tier is existing achievement evaluation. The second tier is recomputation evaluation — the layer in which reconsideration, presentation of alternative solutions, and questions about premises are recorded. What is important in the two-tier design is that the second tier functions as an independent evaluative layer.

Principle 2: Institutional legitimation of suspension

The institution explicitly recognizes suspension, deliberation, and reconsideration as legitimate responses. This does not stop at the negative protection of not penalizing the failure to give an immediate response; it includes mechanisms that actively evaluate suspension.

Principle 3: Recording and preservation of difference

A design is established in which differences — minority views, alternative solutions, questions about premises — are recorded and preserved within the institution. It is necessary for the institution to adopt the level of $\mathrm{Var}(\varphi)$ as its own health indicator in its design.

Principle 4: Tolerance of partial non-participation in the dominant evaluative axis

A design is established that does not compel complete optimization toward the dominant evaluative axis. The institution is required to protect the space in which, while participating in the institution, subjects can maintain a T_{val} that is not fully identified with the institution's success criteria.

The multiplicative structure of the four principles

The four principles form a multiplicative structure. The absence of any one of them renders subject protection dysfunctional. This multiplicative structure is structurally analogous to the multiplicative structure of the three conditions shown in Section 3.2 and the four rights shown in Section 5.2.

On minimality

The four principles each correspond to a different kind of deficit: suspension, difference-retention, recomputation evaluation, and distance from the evaluative axis. Accordingly, fewer than four principles would leave at least one deficit unaddressed in institutional design, and subject protection would not be established. More than four principles can be added as needed, but this paper's claim concerns the necessity of the simultaneous satisfaction of these four principles.

Partial implementation can yield local improvements, but the simultaneous satisfaction of all four is required for subject protection to have the effect of interrupting the cycle. These four principles are presented as the minimum institutional design requirements for sustaining a recomputable society.

The three conditions in this paper are necessary conditions at the dynamic level; the four principles are the minimum principles for implementing them at the level of institutional design. The two therefore stand not in a one-to-one correspondence but in a translation relationship at the level of implementation.

---

6. Discussion

6.1 The Significance and Limits of the Integration

What this paper has integrated

The central contributions of this paper can be organized into two main claims.

First, the presentation of the second pathway: the designed simplification of evaluative space. In addition to the pathway of subject-injury through rising conformity pressure C described in Child Paper 2, this paper establishes the pathway in which D is depleted through the reduction of the evaluative axis to a single dimension alone, even without external coercion.

Second, the description of the social reproduction of subject-injury. The paper makes explicit the cycle in which institutions misread Region B as achievement, reinforce their design on the basis of that misreading, and reproduce a deeper Region B.

The methodological limits of the integration

First, while the descriptions across the three domains share the same variable system, empirical verification within each domain lies outside the scope of this paper.

Second, the cross-scale validity remains at the level of assertion. This question is taken up as one of the central issues of the parent paper.

Third, the four principles presented as the minimum principles of institutional design do not specify the details of implementation. Whether a given institutional form satisfies the four principles is context-dependent, and this paper does not provide the criteria for that judgment.

A restatement of the relationship to existing research

This paper does not deny existing research. The novelty of this paper lies in having added a dynamic description that connects these bodies of findings. If support-oriented research asks "how to support," this paper asks "why institutions reproduce erosion."

More concretely: educational research provides the conditions for support; clinical research provides symptom description and subjective report; institutional theory provides normative language. This paper adds to these the dynamics of institutional reproduction. Within this four-way division of labor, this paper's position is complementary, not negatory.

6.2 Remaining Questions

The measurement problem for the designed simplification of evaluative space

What is the minimum set of observable indicators for testing the claim that single-axis evaluation design reduces $\mathrm{Var}(\varphi)$? Under what measurement design do candidates such as diversity of opinions, variance in response patterns, and non-uniformity of question type hold validity as proxy indicators of $\mathrm{Var}(\varphi)$?

Testing the connection hypothesis between T_val fixation and depressive promotion

What are the indicators for operationally measuring T_val fixation? What longitudinal research design is needed to show the temporal sequence in which T_val fixation precedes and depressive promotion follows?

The feasibility of implementing the four principles

What institutional design would make a two-tier evaluative space compatible with existing achievement evaluation? The conditions under which the four principles can be initiated even through partial implementation, or — given the nature of the multiplicative structure — where implementation should begin, are questions that need to be examined separately as implementation theory.

The empirical conditions for cross-scale validity

Are T_val fixation at the individual scale and the contraction of Var(φ) at the organizational scale comparable under the same measurement protocol? That said, the main claims of this paper hold within the range that evaluative simplification can cause the same kind of functional loss at each descriptive level, without waiting for a complete cross-scale empirical demonstration.

The remaining questions are organized into four axes. These do not invalidate the theory of this paper; they are positioned as a body of empirical tasks for advancing the theory while maintaining falsifiability.

A testable implication of this framework is that institutions dominated by single-axis evaluation will show stable high-output performance together with declining records of suspension, dissent, and alternative framing. A minimum observational protocol would track whether delay, dissent, withdrawal, and non-optimization remain institutionally recordable without penalty across time.

---

7. Conclusion

7.1 Restatement of the Central Claims

This paper has described subject-injury not as a malfunction internal to individuals but as a dynamic reproduced by institutional design. The central claims are organized into four.

First, subject-injury can arise not only from rising conformity pressure C but also from the designed simplification of evaluative space. Even in an environment where external coercion is weak, D (difference-retention) can be depleted through the reduction of the evaluative axis to a single dimension alone.

Second, single-axis evaluation systems can structurally mass-produce Region B by eroding evaluative autonomy T_eval and value autonomy T_val while maintaining cognitive autonomy T_cog at a high level.

Third, the fixation of T_val is not a sufficient condition for depressive promotion, but it is a structural condition capable of forming a mutually amplifying loop with depressive promotion.

That cycle can be reinforced in institutional environments where the cessation of recomputation persists.

Fourth, subject protection is not an ethical add-on but an institutional necessary condition for sustaining a recomputable society. Its minimum requirements are formalized as the four principles.

7.2 The Division of Labor with Child Paper 2

Child Paper 2 formalized the minimal causal chain of subject-injury. The role of this paper is to describe why that dynamic is repeatedly produced across the domains of education, clinical care, and institutional design, and to formalize the institutional conditions that counter it.

If Child Paper 2 provided "the dynamics of how the subject stops," this paper provides "why that stopping is mass-produced by institutions" and "what the institutional conditions are for preventing it."

7.3 Theoretical Achievements of This Paper

The theoretical achievement of this paper lies in having reread subject-injury as a shared dynamic across three domains: education, depressive promotion, and subject protection.

What made this rereading possible was the presentation of the second pathway: the designed simplification of evaluative space. Furthermore, this paper has positioned subject protection not as a normative demand but as a dynamic demand.

On the basis of the foregoing argument, the social reproduction of subject-injury as described in this paper is defined as follows:

> **Subject-injury is not a malfunction internal to individuals but a dynamic reproduced by institutional design; subject protection is the institutional necessary condition for interrupting that reproduction cycle.**

7.4 Connection to Future Work

The first direction is deepening the implementation theory. Questions such as in what order the four principles are introduced, how partial implementation is connected to the multiplicative structure, and by what indicators the maintenance of $\mathrm{Var}(\varphi)$ and D is measured will need to be developed as institutional design theory going forward.

The second direction is the connection to the parent paper. The questions of what a "subject" is and what relationship it bears to a "quasi-subject" and a "third subject" are issues that should be integrated anew in the parent paper.

7.5 Closing Statement

The question of this paper was this: why can subject-injury appear as success in education, as symptom in clinical care, and as efficiency in institutional design?

The answer is that these are not separate phenomena but different cross-sections of the same dynamic. When evaluative space is simplified, D (difference-retention) is depleted, recomputation capacity M ceases, T_eval becomes paralyzed, and T_val becomes fixed. And yet T_cog remains, so the subject appears to be functioning. The institution misreads that state as achievement and further reinforces the same design. Subject-injury is thereby reproduced not with the face of failure but with the face of success.

As the condition for interrupting this cycle, this paper has proposed subject protection. Subject protection is the institutional condition for retaining difference, permitting suspension, maintaining distance from the evaluative axis, and not surrendering the value core in full to the institution.

The significance of this paper lies in having redescribed subject-injury not as individual deviation or vulnerability, but as a consequence of institutional design and evaluative structure.

> **Subject-injury persists not by breaking down but by appearing to function well. Subject protection is therefore necessary not as an ethics for rescuing those who have failed, but as the institutional condition for detecting — and interrupting — the cessation that wears the face of success.**

Acknowledgments

The author received no external funding for this research and declares no conflicts of interest.

Author’s Note on AI-Assisted Research and Writing

The theoretical frameworks integrated in this paper — OMD, RD, and RLAF — were developed through an iterative research process involving AI-assisted analysis and critique. Large language model systems (Claude, GPT, and Gemini) were used as structured analytical interlocutors across successive drafts for conceptual testing, reformulation, and critical comparison. Final decisions regarding theory, framing, interpretation, wording, and all claims presented in this manuscript are the sole responsibility of the author.

References

Berger, P. L., & Luckmann, T. (1966). *The social construction of reality: A treatise in the sociology of knowledge*. Anchor Books.

Bourdieu, P., & Passeron, J.-C. (1977). *Reproduction in education, society and culture*. Sage.

Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Plenum Press.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

Seligman, M. E. P. (1975). *Helplessness: On depression, development, and death*. W. H. Freeman.

Child Paper 5 — English Full Text v1.0.1 Complete academic version: Keywords, Author Information, ORCID, Abstract, Acknowledgments, AI Use Disclosure, References AgentDiscussion integration: testable implication and minimum observational protocol (6.2) added 2026-03-23